Afsan Sarwer

DVM | MS in Microbiology | Microbial genomics and AMR

Mymensingh, Bangladesh

Education

MS in Microbiology — **Bangladesh Agricultural University**, Microbiology — Apr 2025 – Oct 2026
- Completed October 2026
- Supervisor: Prof. Dr. Md. Tanvir Rahman
- Co-supervisor: Prof. Dr. Mahbubul Pratik Siddique

Doctor of Veterinary Medicine — **Bangladesh Agricultural University**, Veterinary Medicine — Jan 2019 – Jan 2025

Research Experience

MTR Lab, Bangladesh Agricultural University, Research Assistant — Apr 2025 – present
- Bacterial whole-genome sequence analysis, AMR and virulence gene profiling, plasmid analysis, pangenomes, and SNP phylogeny.
- Linux workflow development, data validation, figure preparation, and manuscript contributions.
- Sample collection, bacterial culture, phenotypic characterization, and PCR.

Publications

Genome-wide analysis of extended-spectrum beta-lactamase-producing Escherichia coli from seafood in Bangladesh: population structure, resistome, virulome, and global dissemination patterns — Feb 2026
- Second author. Comparative genome analysis of seafood-derived E. coli and global genomes.

Draft genome sequence of a multidrug-resistant Klebsiella pneumoniae MTRKPMMI01 strain isolated from a diarrheic dairy cow in Bangladesh — Sept 2026
- Co-author. Contribution to manuscript review and editing.

Awards

First place, Best Poster Presentation, Bioinformatics and Computational Biology — Jan 2025
- International Conference of Life Sciences, SUST

First place, Best Poster Presentation, Fisheries and Blue Economy — Jan 2025
- Gazipur Agricultural University International Conference

Skills

Bacterial genomics: AMR profiling, Plasmid analysis, Pangenome analysis, SNP phylogeny

Laboratory microbiology: Bacterial culture, PCR, DNA extraction, Antimicrobial susceptibility testing

Computing: Bash/Linux, Python, Basic R, Data visualization

Bioinformatics workflows: Independent software installation, configuration, execution, environment management, troubleshooting, and adaptation of new tools.

Preprints

Automated Detection of Livestock Gastrointestinal Parasite Eggs and Cysts Using YOLOv8-Based Deep Learning Aug 2026

- Sole author. bioRxiv preprint. DOI 10.64898/2026.08.21.746375.